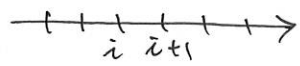
Transfer matrix and mapping of d-dimensional quantum system to (d+1)-dimensional classical.

<u>d=1 classical Ising model</u> → <u>d=0 quantum spin</u>



$$H_{ising} = -J\sum_i \sigma_i\sigma_{i+1} - h\sigma_i$$

$$Z = \sum_{\{\sigma\}} e^{-\beta H_{ising}} = \sum_{\{\sigma\}} e^{\sum_i \beta J\sigma_i\sigma_{i+1} + \beta h\sigma_i} =$$

sum over all configurations

$$= \sum_{\{\sigma\}} e^{\beta J\sigma_1\sigma_2 + \beta h\sigma_1 + \beta J\sigma_2\sigma_3 + \beta h\sigma_2 + \dots} =$$

$$= \sum_{\{\sigma\}} V(\sigma_1,\sigma_2)\, V(\sigma_2,\sigma_3) \dots V(\sigma_N,\sigma_{N+1})$$

assume PBC so that $\sigma_{N+1} = \sigma_1$

Thus Z is represented as a product of 2×2 matrices V

$$V(\sigma,\sigma') = \exp\left[\beta J\sigma\sigma' + \tfrac{1}{2}\beta h(\sigma+\sigma')\right]$$

Its elements are
$$\langle +|V|+\rangle = e^{\beta J+\beta h}$$
$$\langle +|V|-\rangle = e^{-\beta J} = \langle -|V|+\rangle$$
$$\langle -|V|-\rangle = e^{\beta J-\beta h}$$

Think of classical configuration space of N-spins as a <u>temporal</u> evolution of the quantum two-state system (so that site index i becomes i-th slice along the time axis)

$$|\sigma_{i+1}\rangle = \hat{V}|\sigma_i\rangle = e^{-a\hat{H}_{TM}}|\sigma_i\rangle$$ and $a = \Delta t$ is the step on time-direction

Note $\hat{H}_{TM} \neq \hat{H}_{ising}$

Rather, $\hat{H}_{TM}$ is an effective evolution operator

Transfer matrix V "transfers" spin state at site $\underline{i}$ to that at site $\underline{i+1}$.

(Let for simplicity $h=0$ below) $\Rightarrow$ $\langle +|V|+\rangle = \langle -|V|-\rangle = e^{\beta J}$

$$\langle -|V|+\rangle = \langle +|V|-\rangle = e^{-\beta J}$$

$$V = \begin{pmatrix} e^{\beta J} & e^{-\beta J} \\ e^{-\beta J} & e^{\beta J} \end{pmatrix} = e^{\beta J}\,\hat{\sigma}_0 + e^{-\beta J}\,\hat{\sigma}_1$$

Pauli matrices $\hat{\sigma}_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \hat{\sigma}_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \ldots$

$$\Rightarrow V = \exp(c_0\,\hat{\sigma}_0 + c_1\,\hat{\sigma}_1) = \underbrace{e^{c_0}}_{\tilde{c}}\,\hat{\sigma}_0\; e^{c_1\hat{\sigma}_1}$$

$$e^{c_1\hat{\sigma}_1} = \sum_{n=0}^{\infty} \frac{(c_1\hat{\sigma}_1)^n}{n!} = \sum_{n=\text{even}} \frac{c_1^n}{n!} + \sum_{n=\text{odd}} \frac{c_1^n}{n!}\,\hat{\sigma}_1 =$$

$$(\hat{\sigma}_1)^n = \begin{cases} 1, & n=\text{even} \\ \hat{\sigma}_1, & n=\text{odd} \end{cases}$$

$$= \left(1 + \frac{c_1^2}{2!} + \frac{c_1^4}{4!} + \ldots\right) + \hat{\sigma}_1\left(\frac{c_1}{1!} + \frac{c_1^3}{3!} + \ldots\right) =$$

$$= \cosh c_1 + \hat{\sigma}_1 \sinh c_1$$

$$\Rightarrow V = \tilde{c}(\mathrm{ch}\, c_1 + \hat{\sigma}_1\, \mathrm{sh}\, c_1) = e^{\beta J}\hat{\sigma}_0 + e^{-\beta J}\,\hat{c}_1$$

$$\left.\begin{aligned} e^{\beta J} &= \tilde{c}\,\mathrm{ch}\, c_1 \\ e^{-\beta J} &= \tilde{c}\,\mathrm{sh}\, c_1 \end{aligned}\right\} \mathrm{th}(c_1) = e^{-2\beta J}$$

$$(\tilde{c})^2(\mathrm{ch}^2 c_1 - \mathrm{sh}^2 c_1) = e^{2\beta J} - e^{-2\beta J}$$

$$e^{c_0} = \tilde{c} = \sqrt{e^{2\beta J} - e^{-2\beta J}} = \sqrt{2\,\mathrm{sh}(\beta J)}$$

Thus $-a\hat{H}_{TM} = \cancel{\ln\tilde{c}(\mathrm{ch}\, c_1 + \mathrm{sh}\, c_1\,\hat{\sigma}_1) = \ln[\tilde{c}\,\mathrm{ch}\, c_1(1 + \mathrm{th}(c_1)\,\hat{\sigma}_1)]}$

$$c_0 + c_1\hat{\sigma}_1$$

$$\hat{H}_{TM} = -\frac{1}{a}\left\{\underbrace{\frac{1}{2}\ln[2\,\mathrm{sh}(\beta J)]}_{c_0} + \underbrace{\frac{1}{2}\ln\left(\frac{1+e^{-2\beta J}}{1-e^{-2\beta J}}\right)}_{c_1}\hat{\sigma}_1\right\}$$

"High-T" limit: $\beta J = \frac{J}{k_B T} \ll 1 \Rightarrow e^{-2\beta J} = 1 - 2\beta J$

$$\hat{H}_{TM} = -\frac{1}{a}\,\frac{1}{2}\ln\left(\frac{2}{\beta J}\right)\hat{\sigma}_1$$

"Low-T" limit: $\beta J \gg 1 \Rightarrow e^{-\beta J} \ll 1 \Rightarrow$

$$\hat{H}_{TM} = -\frac{1}{a}\,\frac{1}{2}\left(2e^{-2\beta J}\right)\hat{\sigma}_1 = -\left(\frac{e^{-2\beta J}}{a}\right)\hat{\sigma}_1 .$$

The quantum Hamiltonian $\hat{H}_{TM}$ is finite, in the limit of $\underline{a \to 0}$, only when the combination $(a e^{2\beta J} = \text{const})$ is held constant.

Now, let's look back on the partition function via V:

$$Z = \sum_{\{\sigma\}} \langle\sigma_1|V|\sigma_2\rangle\langle\sigma_2|V|\sigma_3\rangle \ldots \langle\sigma_N|V|\sigma_1\rangle =$$

$$= \sum_{\sigma_1=\pm}\sum_{\sigma_2=\pm}\sum_{\sigma_3=\pm} \langle\sigma_1|V|\sigma_2\rangle\langle\sigma_2|V|\sigma_3\rangle \ldots \langle\sigma_N|V|\sigma_1\rangle$$

$$= \left(\text{but } \sum_{\sigma_2=\pm} \langle\sigma_1|V|\sigma_2\rangle\langle\sigma_2|V|\sigma_3\rangle = \langle\sigma_1|V^2|\sigma_3\rangle\right) =$$

$$= \sum_{\sigma_1=\pm} \langle\sigma_1|V^N|\sigma_1\rangle = \mathrm{Tr}\left(\hat{V}^N\right).$$

Let U be diagonalizing matrix so that $V = U D U^{-1}$, $U U^{-1} = \hat{1}$, where $D = \begin{pmatrix} \lambda_+ & 0 \\ 0 & \lambda_- \end{pmatrix}$ is the matrix of eigenvalues, $\lambda_+ > \lambda_-$.

$$\mathrm{Tr}(V^N) = \cancel{\mathrm{Tr}((UDU^{-1})^N} \cancel{= \mathrm{Tr}(UDD)} \; \mathrm{Tr}\left(U U^{-1} V^N\right) =$$

$$= \mathrm{Tr}\big(\underbrace{U V^N U^{-1}}_{U V^{N-1} \underbrace{U^{-1} U}_{D} V U^{-1}}\big) = \mathrm{Tr}\left(U V^{N-1} U^{-1} \cdot D\right) = \mathrm{Tr}\left(D^N\right) = \lambda_+^N + \lambda_-^N .$$

$$\mathrm{Tr}\left(D^N\right) = \mathrm{Tr}\begin{pmatrix} \lambda_+ & 0 \\ 0 & \lambda_- \end{pmatrix}^N = \mathrm{Tr}\begin{pmatrix} \lambda_+^N & 0 \\ 0 & \lambda_-^N \end{pmatrix}$$

The eigenvalues are

$$\det\left(V-\lambda\hat{1}\right)=0=\det\begin{pmatrix} e^{\beta J}-\lambda & e^{-\beta J} \\ e^{-\beta J} & e^{\beta J}-\lambda \end{pmatrix}=\left(e^{\beta J}-\lambda\right)^2-\left(e^{-\beta J}\right)^2=0$$

$$\Rightarrow e^{\beta J}-\lambda=\pm e^{-\beta J} \Rightarrow \lambda=e^{\beta J}\mp e^{-\beta J}=\begin{cases} 2\,sh(\beta J)=\lambda_- \\ 2\,ch(\beta J)=\lambda_+ \end{cases}$$

Since $\lambda_+=2ch(\beta J)>\lambda_-=2sh(\beta J)$, $Z=\lambda_+^N+\lambda_-^N\to\lambda_+^N$ for $N\gg 1$.

$$\Rightarrow \beta F=-\ln Z=-N\ln(\lambda_+)=-N\ln\left(e^{\beta J}+e^{-\beta J}\right).$$

is indeed the free energy of N-spin ~~[illegible]~~ Ising chain.

The quantum connection with $\hat{H}_{TM}$ allows to understand the absence of spontaneous symmetry breaking (ordering) at any finite temperature β^{-1} as ~~[illegible]~~ a quantum effect — an equivalent quantum spin, evolving in time β, never ceases to tunnel $\Rightarrow$ there can not be symmetry breaking.

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$d=1$ transverse field Ising chain $\longrightarrow$ $d=2$ classical Ising.

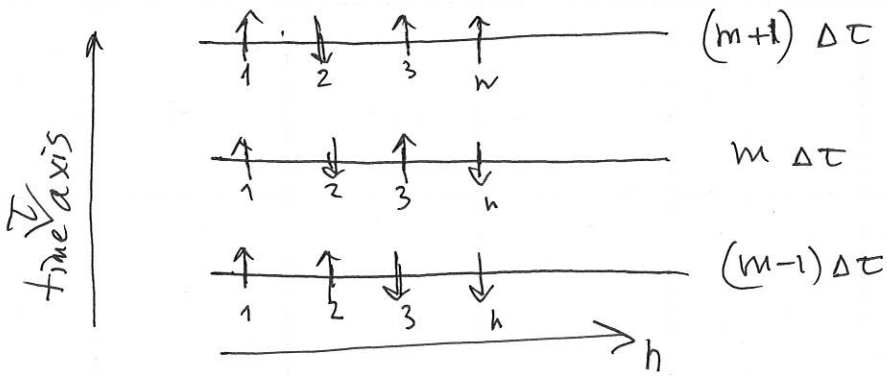
$$Z_Q = \mathrm{Tr}\left\{ e^{-\beta \hat{H}_{TFIM}} \right\}$$

$$\hat{H}_{TFIM} = \hat{H}_1 + \hat{H}_2 = -J \sum_n \sigma_n^z \sigma_{n+1}^z - Jg \sum_n \sigma_n^x$$

Trotter decomposition: $\beta = \Delta\tau \cdot N$, insert resolution of unity at every time slice

$$Z_Q = \sum_{\{S_i(\tau)\}} \langle \{S_i(N)\} | e^{-\Delta\tau \hat{H}} | \{S_i(N-1)\} \rangle \ldots \langle S_i(k) | e^{-\Delta\tau \hat{H}} | S_i(k-1) \rangle \ldots \langle \{S_i(1)\} | e^{-\Delta\tau \hat{H}} | \{S_i(N)\} \rangle$$

where $\{S_i(m)\}$ is the spin configuration at $(S_i(m) = \pm 1)$ time step $m\Delta\tau = \frac{m}{N}\beta$, i = coordinate along the chain.



Since $\hat{H}_1, \hat{H}_2$ do not commute $e^{-\Delta\tau(\hat{H}_1+\hat{H}_2)} \approx e^{-\Delta\tau \hat{H}_1} e^{-\Delta\tau \hat{H}_2} + O(\Delta\tau^2)$

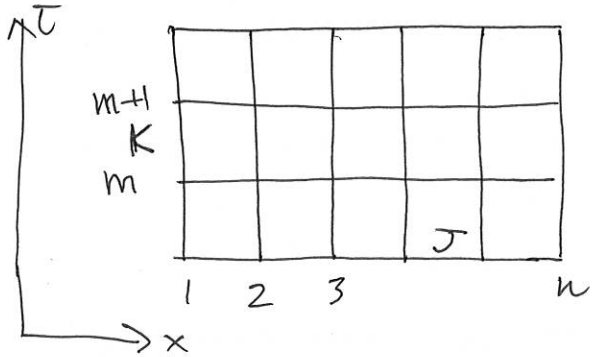
$$\langle \{S_i(m)\} | e^{-\Delta\tau \hat{H}_2} \underbrace{e^{-\Delta\tau \hat{H}_1}}_{e^{+\Delta\tau J \sum_n \sigma_n^z \sigma_{n+1}^z}} | \{S_i(m-1)\} \rangle =$$

$$e^{+\Delta\tau J \sum_n \sigma_n^z \sigma_{n+1}^z} | \{S_i(m-1)\} \rangle = e^{\Delta\tau J \sum_n S_i(m-1) S_{i+1}(m-1)} \cdot | \{S_i(m)\} \rangle$$

$$= e^{\frac{\beta J}{N} \sum_n S_i(m-1) S_{i+1}(m-1)} \cdot \underbrace{\langle \{S_i(m)\} | e^{\Delta\tau J g \sum_n \sigma_n^x} | \{S_i(m-1)\} \rangle}_{\text{same as for single spin}}$$

$$e^{K \sum_n \sum_m S_n(m-1) S_n(m)} \; ; \; e^{-2K} = \mathrm{th}\left(\frac{\beta J g}{N}\right)$$

$\Rightarrow Z_Q$ is described by rectangular 2d ~~system~~ classical Ising model



$$Z_Q = \sum_{S_i(m)=\pm 1} \exp \sum_{i,m} \left(\frac{\beta J}{N} S_i(m) S_{i+1}(m) + K S_i(m) S_i(m+1) \right)$$

x-coupling: $\frac{J\beta}{N}$ $(= K_1)$

y-coupling: $K = \frac{1}{2} \ln\left(\frac{N}{\beta J g}\right)$ $(= K_2)$

2d Ising with couplings K_1 and K_2 (along x and y directions) has phase transition at T_c given by

$$\operatorname{sh}(2K_1) \operatorname{sh}(2K_2) = 1$$

For $K_1 \ll 1$, $K_2 \gg 1$ we have then: $2K_1 \frac{1}{2} e^{2K_2} = 1$

$$\Rightarrow \frac{J\beta}{N} e^{\ln\left(\frac{N}{\beta J g}\right)} = 1$$

$$\frac{J\beta}{N} \frac{N}{\beta J g} = \frac{1}{g} = 1$$

$\Rightarrow$ classical (finite T_c) transition in $d=2$ Ising =

= quantum ($T=0$) transition at $g_c = 1$.